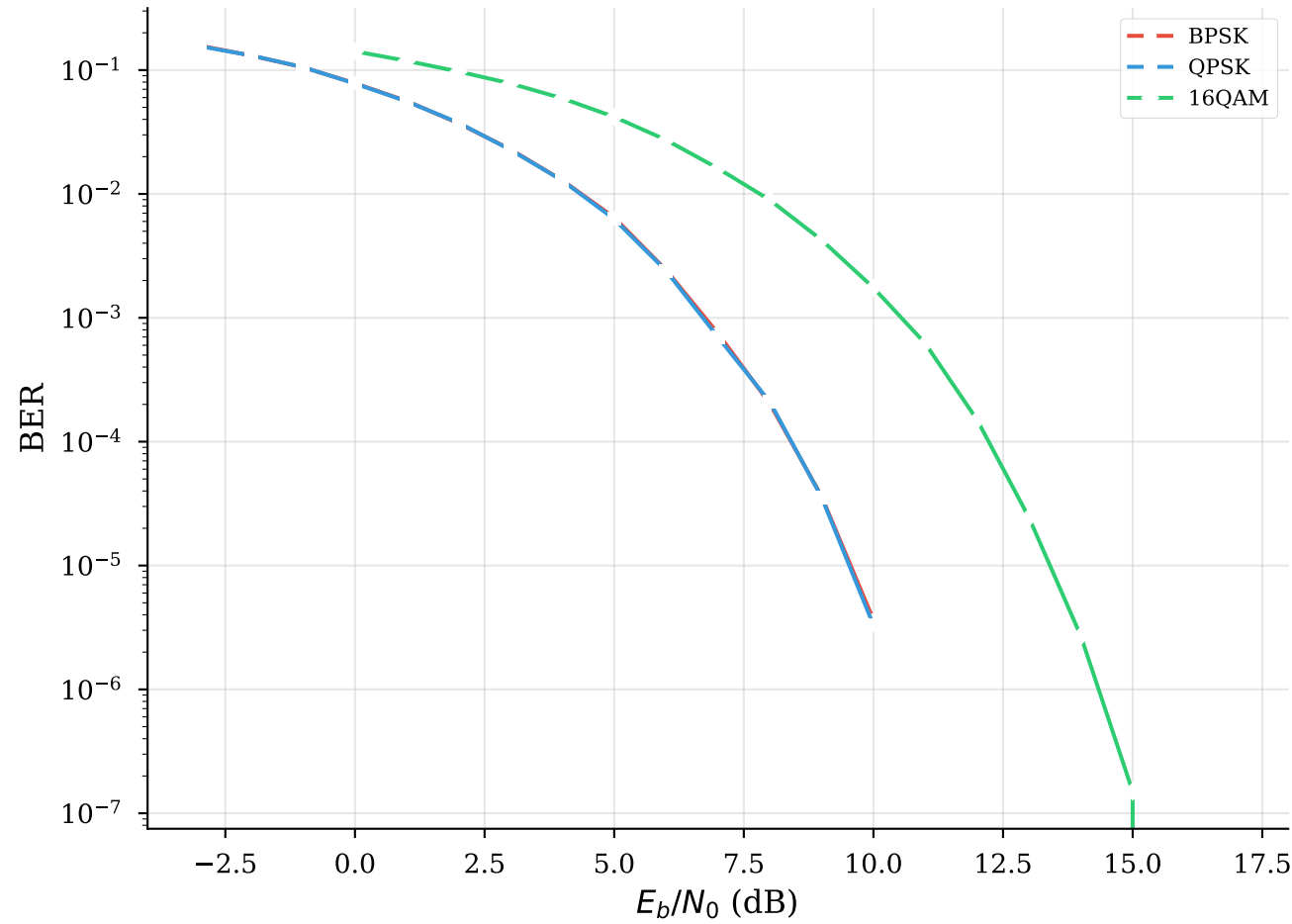
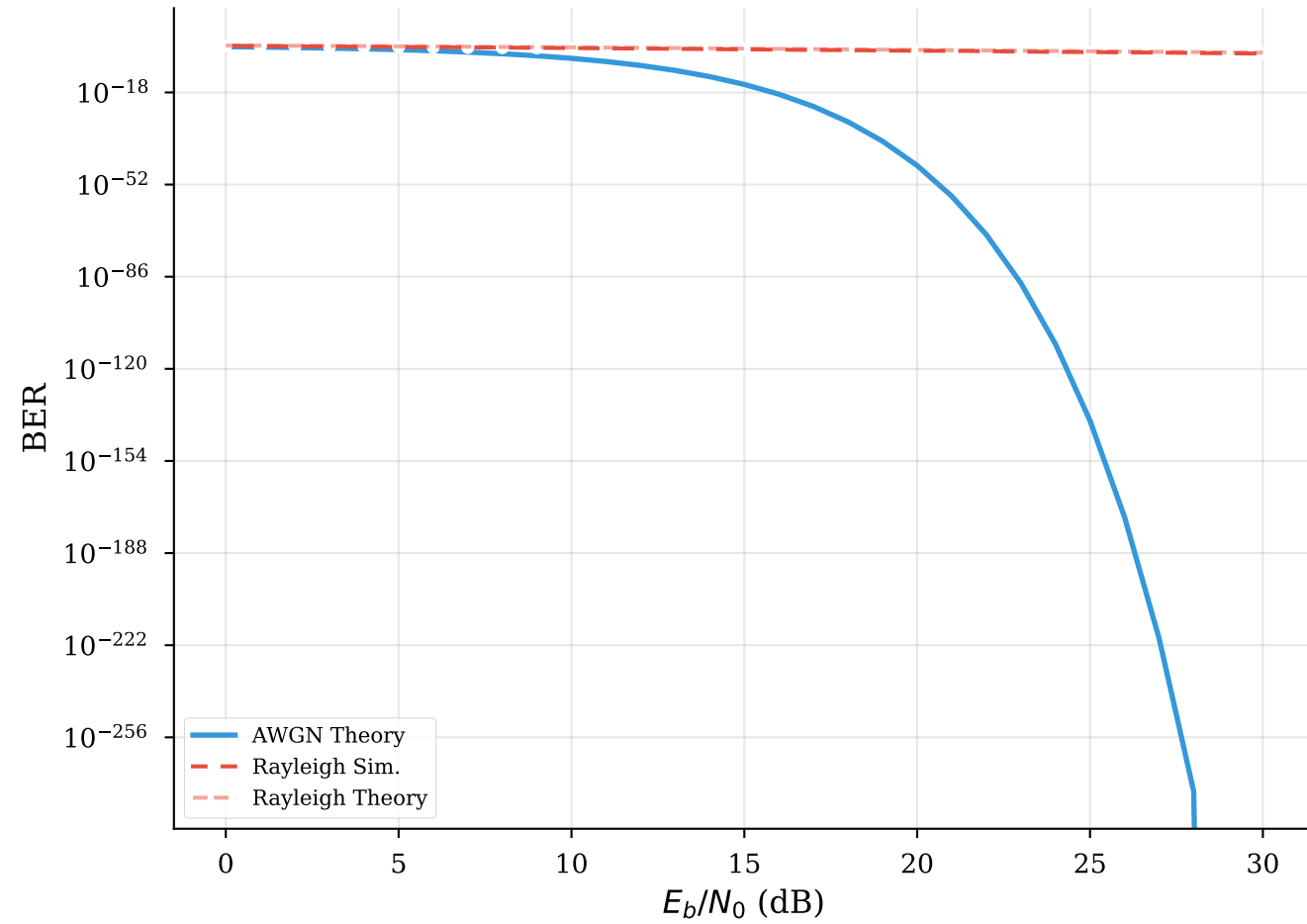


Communication System Simulation: Comprehensive Performance Analysis

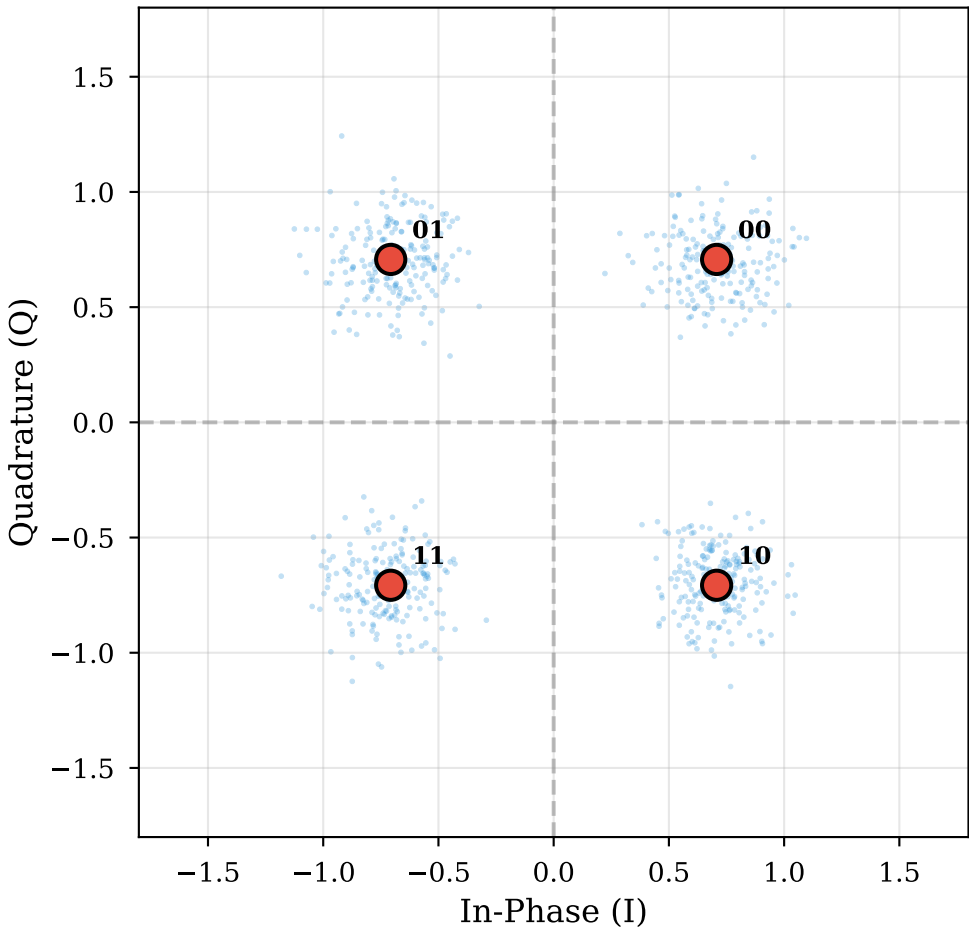
(a) Multi-Modulation BER (AWGN)



(b) AWGN vs Rayleigh Fading



(c) QPSK Constellation (with noise samples)



(d) Simulation Parameters Summary

Parameter	BPSK	QPSK	16QAM
Bits/Symbol	1	2	4
Spectral Eff.	1 bps/Hz	2 bps/Hz	4 bps/Hz
BER= 10^{-3} at	6.8 dB	6.7 dB	10.5 dB
Channel	AWGN / Rayleigh	AWGN	AWGN
Min Errors	100	100	100
Total Bits Processed	$>10^6$	$>10^6$	$>10^7$